

Electricity in the Home

KEYWORDS

Earth

Fuse

Live

Neutral



Queen
Elizabeth
School

Learning objectives

- Recall that domestic supply in the UK is a.c at 50HZ and 230V
- Describe the main features of Live, Earth and Neutral wires

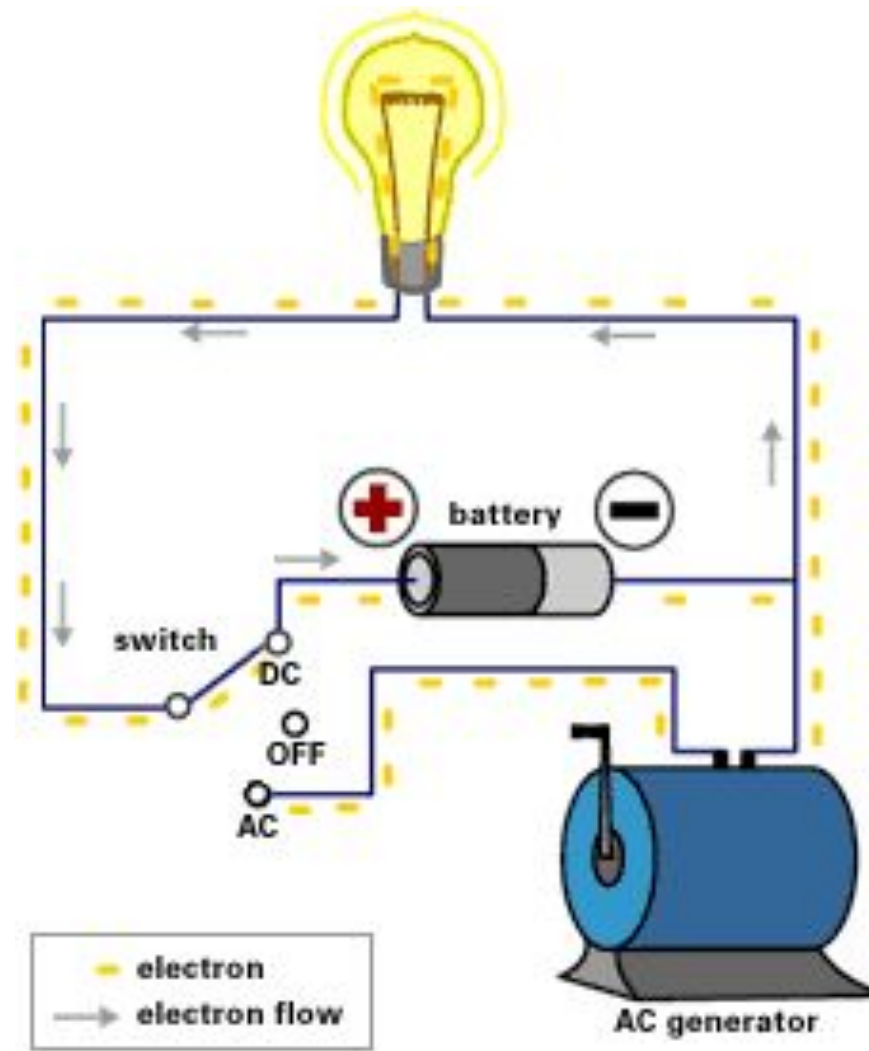
Reproduction rights obtainable from
www.CartoonStock.com



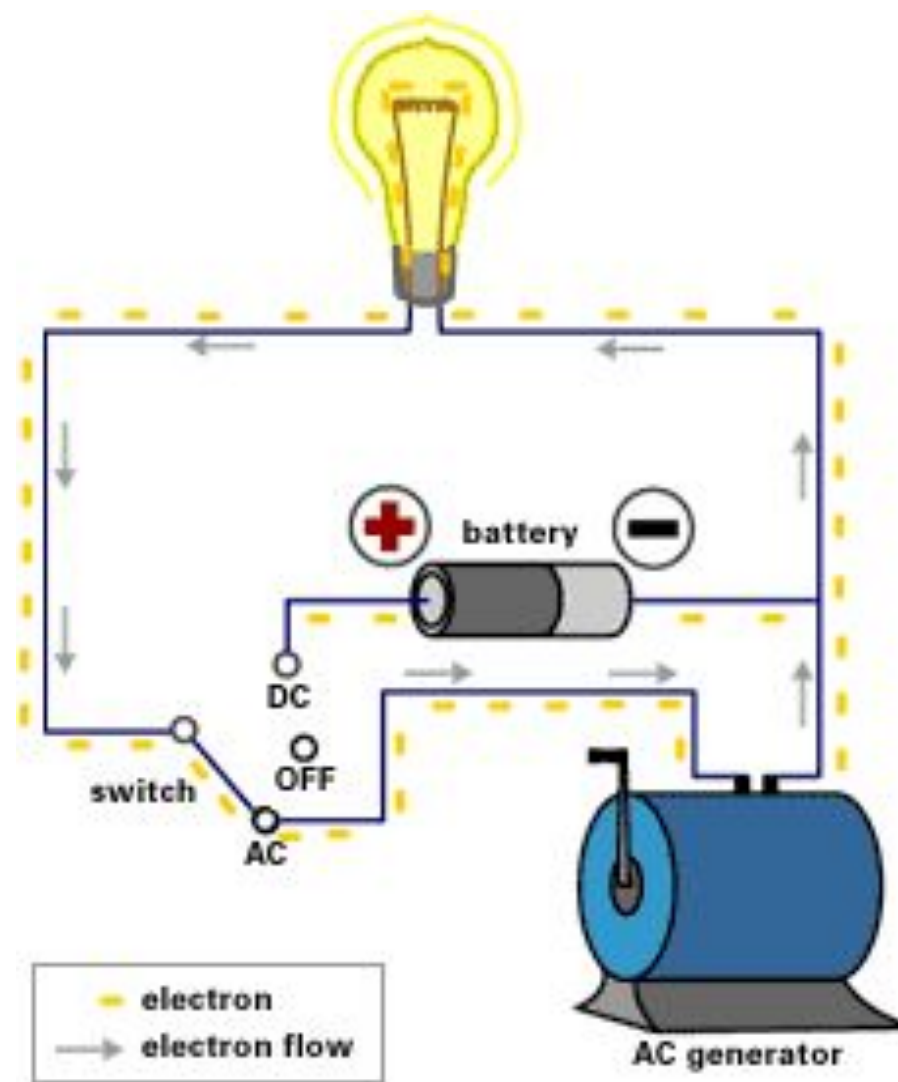
Search ID: ksmn1881

"Have you been trying to fix this
yourself?"

This is called **direct current**.
The electrons flow in one direction and can be controlled by cells or batteries.



You may have noticed the letters a.c. on the power pack. This is **alternating current**. The electrons rapidly change direction. Alternating current can be provided by generators.

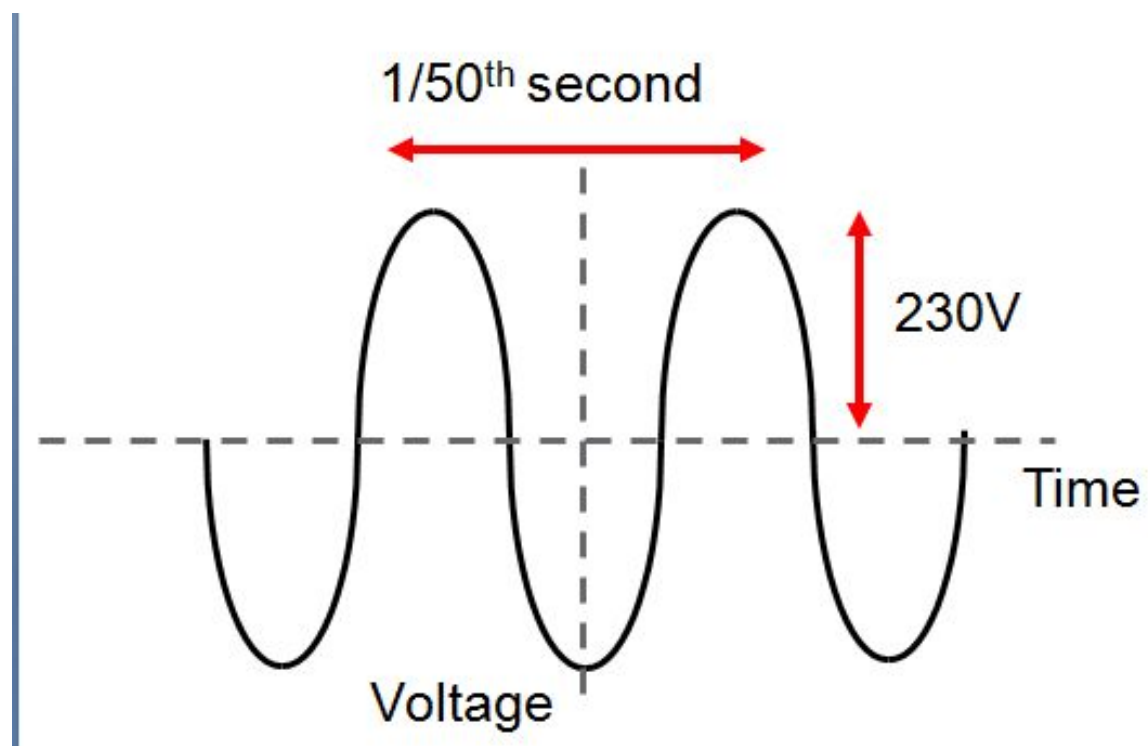


Listen carefully . . .

The hum you hear is the **alternating current** of the mains voltage (from the strip lights).

Mains voltage has a potential difference of **230V** and a frequency of **50Hz**. This means the electrons change direction 50 times a second.





Fill in the missing words

Direct current is where the electrons move in _____ direction.

It is produced by _____.

In an _____ current, the electrons rapidly change
_____.

_____ has an alternating current.

The mains voltage is _____ and has a _____
of 50Hz.

This means that the _____ change direction 50 times
every _____.

one alternating second mains electricity
electrons batteries direction 230V
frequency

CONSOLIDATION

Fill in the missing words.

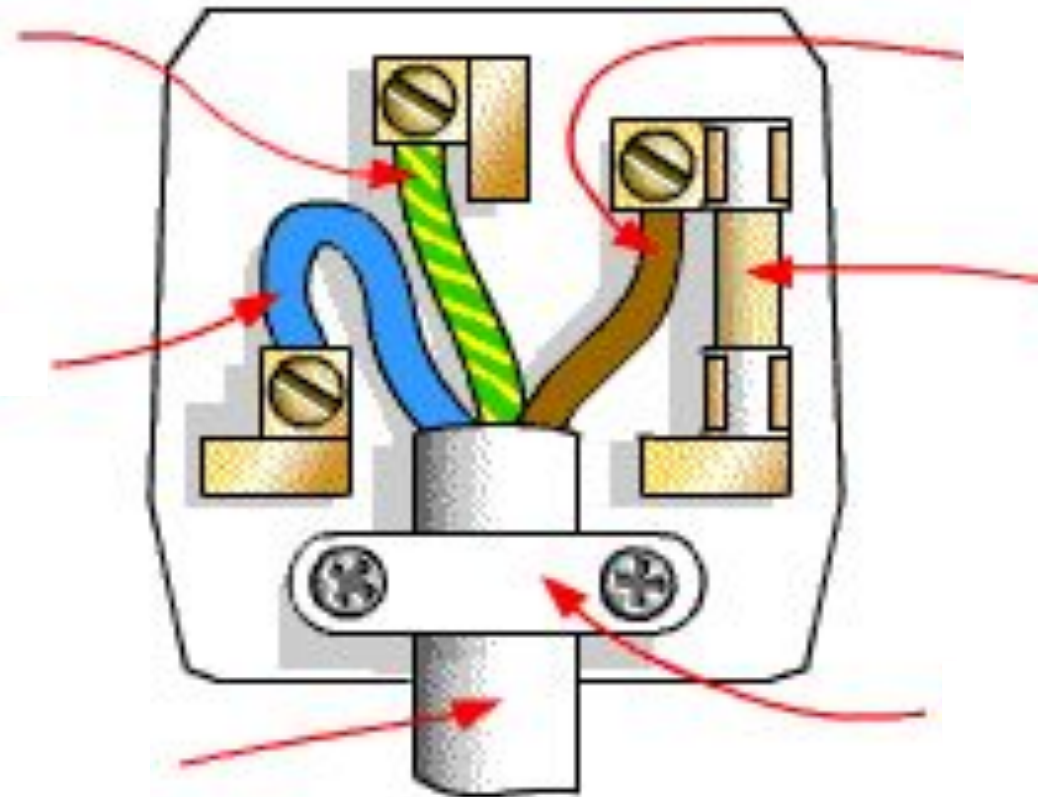
Direct current is where the electrons move in one direction.
It is produced by batteries.

In an alternating current, the electrons rapidly change direction.

Mains electricity has an alternating current.

The mains voltage is 230V and has a frequency of 50Hz.

This means that the electrons change direction 50 times every second.



ACTIVATION

Plugs

Use the information you have learned so far in order to try and wire your plug.

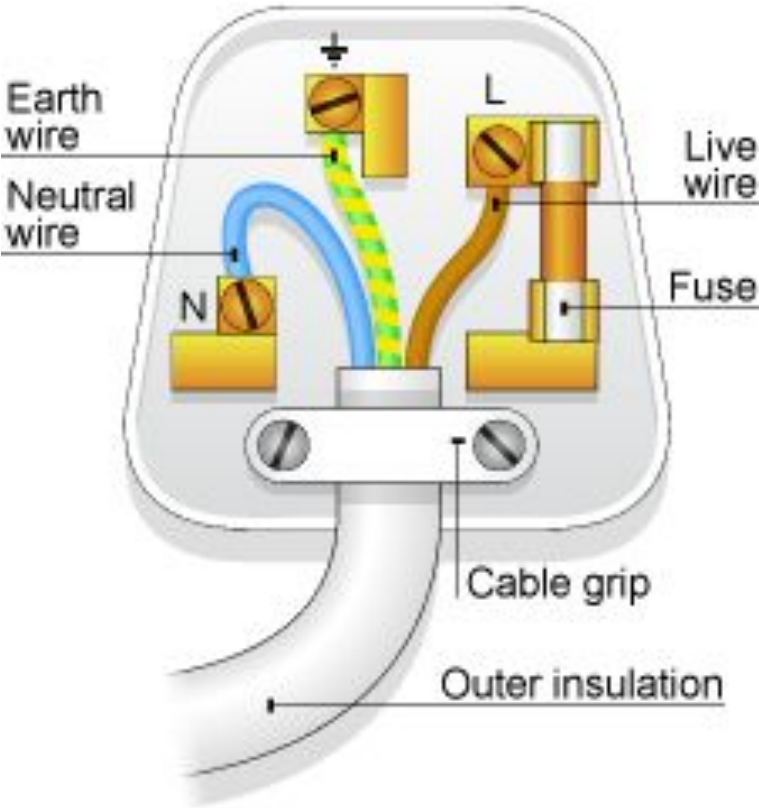
Wire	Insulation Colour	Location
LIVE	BRown	BR – bottom right
NEUTRAL	BLue	BL – bottom left
EARTH	Green/Yellow	Centre

I will be able to describe the main features of live, earth and neutral wires

08:00

Plugs

Use the information you have learned so far in order to try and wire your plug.



Wire	Insulation Colour	Location
LIVE	BRown	BR – bottom right
NEUTRAL	BLue	BL – bottom left
EARTH	Green/Yellow	Centre

ACTIVATION

I will be able to describe the role of a fuse

The role of a fuse

How does a fuse work?

What is the rule for selecting the right fuse?

ACTIVATION



I will be able to describe the role of a fuse

The role of a fuse

Fuses work by having a wire which will melt when the current exceeds a certain value, this will break the circuit.

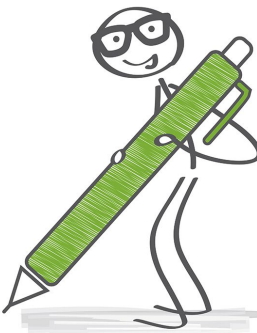
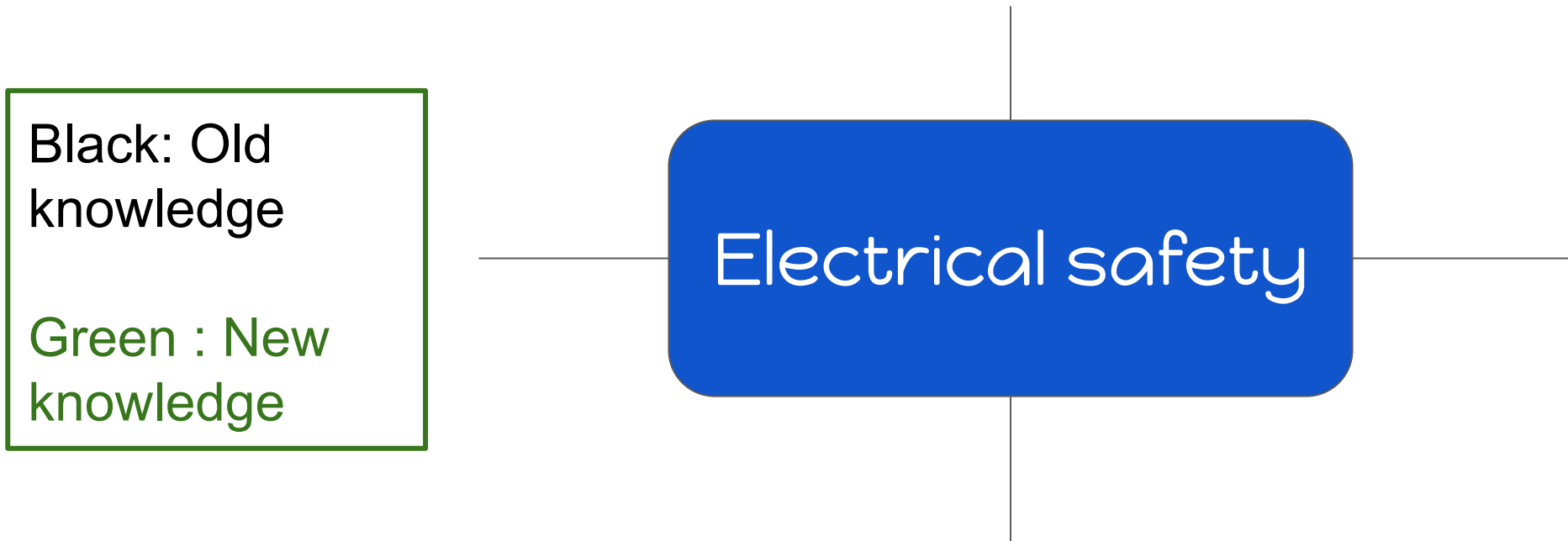


ACTIVATION

And Finally ...

Electrical safety in the home

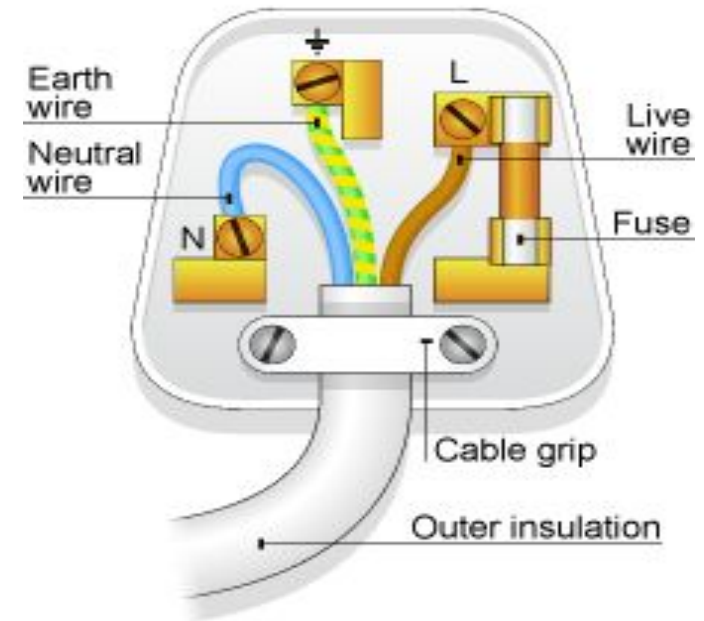
Add new knowledge to your original spider diagram:



Connecting a three pin plug

The features of a plug are:

- **The case** is made from tough plastic or rubber, because these materials are good electrical **insulators**.
- The **three pins** are made from brass, which is a **good conductor** of electricity.
- There is a ***fuse*** between the live terminal and the live pin.
- **The fuse breaks the circuit if too much current flows.**
- The cable is secured in the plug by a **cable grip**. This should grip the cable itself, and not the individual wires inside it.
-



Colour codes of the three wires inside the cable.

- | | |
|----------------------------|-----------|
| • colour | • wire |
| • blue | • neutral |
| • brown | • live |
| • green and yellow stripes | • earth |

Why we get an electric shock when we touch a faulty appliance.

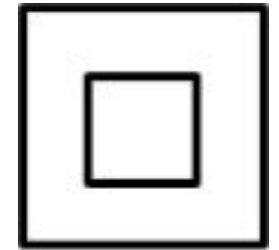
- There is no potential difference between our bodies and the earth because our bodies are at earth potential.
- If the casing of a faulty appliance becomes live, the pd between the metal casing and earth is 230V.
- If a person touches the faulty appliance, there is a pd between the person and the metal casing of 230V. There is a complete circuit and current passes through the person's body to earth so the person receives an electric shock

Why we don't get an electric shock if the appliance is earth

- If the appliance is earthed, it is connected to the earth or ground. Any charge in it flows safely down to the ground through the low resistance earth wire.
- The person is stopped from receiving an electric shock if they touch the live case of a faulty appliance.

How the Live, Neutral and Earth wires work

- Mains electricity comes from a power station.
- The live and neutral wires connect a house to a power station.
-
- **LIVE WIRE**- It carries high pd into and around the house.
- The **FUSE** is always connected to this wire.
- If the current becomes higher than the fuse's rating then the fuse will melt and switch off the circuit.
-
- **NEUTRAL WIRE**-
- It provides a return path to the local substation.
- It is **earthed**.
- It carries no current until an appliance is connected.



Symbol for
double
insulation

A **Double insulated appliance** does not need an earth connection. It has a plastic case with no electrical connections to it so the case cannot become live.

THE EARTH WIRE

- It is a safety wire.
- It is connected to the metal case of an appliance.
- It prevents an appliance becoming charged if touched by a live wire.
- It provides a low resistance path to the ground.
- There is normally no current in it.
-

What is a circuit-breaker?

- It is a resettable fuse. It is used in place of the fuse in the main fuse box. This is because it can be reset at the flick of a switch.

Your turn

- **Questions**
- Answer questions 1-4 p66-67 (ALL)
- **Higher** Answer questions 1-6 p66-67

CONSOLIDATION